

Fine-Tune It the Way You Trained It: Field-Target Reinforcement Learning for One-Step Policies

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Abstract

One-step generative policies distill multimodal demonstration data into a single network evaluation, but fine-tuning them with reinforcement learning is structurally awkward: their likelihoods are intractable, ruling out policy-gradient ratios, and backpropagating a learned critic through the policy ties the size of every update to the magnitude of the critic’s action-gradient — a channel through which an overfit critic drives the policy off the data manifold. We show in a controlled two-regime experiment that this failure is catastrophic precisely when critic data is scarce relative to action dimension, that behavior-cloning penalties do not prevent it, and that it disappears when the update is expressed as bounded particle transport instead of a backpropagated loss. Building on this, we propose Porygon: reinforcement fine-tuning of a one-step drifting policy with the same field-target regression primitive that pretrained it. A frozen base proposes action particles; a residual head is regressed onto targets displaced by a clipped critic-gradient reward field and a restoring anchor field, yielding a pointwise trust region in action space. On robomimic, Porygon lifts a drifting base from 0.877 to 0.99 (can, surpassing the flow-matching + DICE-RL pipeline at 0.957) and to 0.92–0.94 (square), and ablations over lower-resolution reward fields (top- k , exponentially tilted) map exactly when the critic’s gradient — versus only its ranking — should be trusted.

1 Introduction

One-step generative policies are an appealing endpoint for imitation learning: a drifting model (Deng et al., 2026) captures a multimodal demonstration distribution in a single network evaluation — diffusion-quality action distributions at deployment-friendly cost. It is trained by a simple primitive: repeatedly transport a population of its own samples along a velocity field toward the data, and regress the network onto the transported targets. This paper is about what that primitive buys after pretraining. Like any behavior-cloned policy, they plateau at the quality of their demonstrations, and the natural next step is reinforcement-learning fine-tuning with sparse task reward. That step, however, is structurally awkward for exactly this policy class.

The two standard fine-tuning routes both break. Likelihood-ratio methods (PPO/GRPO-style) require $\log \pi(a | s)$, which a one-step noise-to-action map does not expose; the chain-MDP reformulations that rescue multi-step diffusion policies (Ren et al., 2025) have no analog when there is only one step. The remaining route — learn a critic and backpropagate $-Q$ through the actor, as the state-of-the-art residual fine-tuner DICE-RL (Anonymous, 2026c) does — silently ties the size of every policy update to the magnitude of the critic’s action-gradient. A learned critic is trained to rank actions near its data; its gradient field off-manifold is unconstrained (Fujimoto et al., 2018; Anonymous, 2025c). In a controlled toy (Section 3.3), this failure is total: with high-dimensional actions and scarce critic data, the backpropagated update drives true reward to zero while its internal Q -estimate rises by four orders of magnitude — and a BC penalty does not save it, because a loss-space penalty must out-shout an unbounded gradient. Yet in the complementary regime, where the critic

generalizes, the same gradient is the best available signal and value-only alternatives waste it. A useful update rule must be safe in the first regime without discarding the gradient in the second.

The lesson we draw is blunt: a learned critic is a judge, and handing a judge a compass — letting its gradient magnitude set the step — is what breaks. But discarding the compass everywhere, as critic-as-judge methods do (Peng et al., 2019; Wang et al., 2020), wastes the best signal available in the benign regime. The resolution is to change not the signal but the form of the update — and the policy class itself already contains the right form. A drifting model is pretrained by field-target regression: transport samples along a velocity field, regress the network onto the transported targets. Porygon fine-tunes with the very same primitive. We call the resulting method Porygon. A frozen base proposes action particles through a residual head; reward enters as a velocity field V_Q — by default the normalized critic gradient — which is clipped pointwise in action space and combined with a restoring anchor field before the residual is regressed onto the displaced targets. The trust region thus lives where the damage occurs: the policy provably moves at most δ per update at every particle, regardless of critic pathology, a guarantee parameter-space gradient clipping cannot give. No likelihood is ever needed, and in the small-step limit the update provably recovers the backpropagated direction (Proposition 1) — the safety is free.

Because reward enters only as a field, the critic can also be consumed at lower resolution when its gradients cannot be trusted: transport toward critic-ranked top- k particles, or toward the cloud’s own $\exp(Q/\tau)$ -tilted density. Our experiments treat this as an explicit dial and map its regimes. On robomimic, where the critic generalizes, the gradient-field method dominates: from a drifting base at 0.877, Porygon reaches 0.99 on can — above the flow-matching + DICE-RL state-of-the-art pipeline at 0.957 — and 0.92–0.94 on square across seeds, while the value-only sources trail far behind (~ 0.6 on square). [TODO: add square baseline comparison once the matched-budget FM and control runs complete] On the eight hardest LIBERO-90 tasks, where the critic is weaker, the ordering compresses exactly as the toy predicts, with ranking-based transport matching the gradient source. [TODO: finalize hard-8 sentence once chains complete.]

Our contributions:

- A field-target RL update for one-step policies (Section 4): reward, trust region, and anchoring expressed as composable velocity fields, with a pointwise action-space movement bound and a dead-zone restoring anchor; likelihood-free by construction and a strict generalization of the backpropagated residual actor.
- A controlled diagnosis of critic exploitation (Section 3.3): a two-regime toy showing the failure is real, regime-dependent, and not fixed by BC penalties — motivating trust regions in action space rather than loss space.
- Evidence that the form is what wins (Section 5): under matched budgets and family-shared hyperparameters, Porygon beats the backpropagated update on the same base with the same reward signal, lifts a weaker one-step base past the stronger flow-matching + DICE-RL pipeline, and the V_Q -resolution ablation confirms the two-regime picture on real benchmarks.

2 Related Work

RL fine-tuning of generative robot policies. Policy-gradient fine-tuning of vision-language-action models with sparse task reward is now standard for autoregressive token policies (Anonymous, 2025d), but transfers poorly to continuous generative heads: likelihood ratios are intractable for diffusion and flow policies, and workarounds either reformulate the denoising chain as an MDP (Ren et al., 2025; Black et al., 2024) — a construction with no analog for one-step generators — or avoid likelihoods entirely (Park et al., 2025; Anonymous, 2025a; Mark et al., 2024). Closest to our setting, DICE-RL (Anonymous, 2026c) fine-tunes a frozen one-step distillation of a flow-matching policy by training a residual head with a backpropagated critic and a filtered BC anchor; it is our primary baseline and the control

for our actor-update ablation, since we keep its critic, residual parameterization, and data pipeline unchanged.

Critic-as-compass versus critic-as-judge. That a learned Q -function’s action-gradient cannot be trusted is a recurring diagnosis: overestimation and narrow spurious peaks in TD3 (Fujimoto et al., 2018), extrapolation error in BCQ (Fujimoto et al., 2019), and spurious local optima that make sample-and-select beat gradient ascent (Anonymous, 2025c). One response treats the critic only as a judge: weighted regression and sample-based improvement (AWR, CRR, MPO, IDQL, QT-Opt) (Peng et al., 2019; Wang et al., 2020; Abdolmaleki et al., 2018; Hansen-Estruch et al., 2023; Kalashnikov et al., 2018) tilt the policy toward $\pi \cdot \exp(Q/\tau)$ using only Q -values. Our method spans this spectrum rather than picking a side: the update form is judge-safe (bounded transport toward targets), while the reward field inside it can still be the analytic gradient where it is trustworthy — and our ablations map exactly when each choice wins.

Particle transport as a policy update. Regressing a sampler onto displaced particles is the amortized-SVGD idea of Soft Q-Learning (Haarnoja et al., 2017), with kernel repulsion preserving diversity as in SVPG (Liu et al., 2017); drifting models (Deng et al., 2026) make the same operator the pretraining objective of a one-step generator. Drifting Field Policy (Anonymous, 2026b) is the closest prior method: it also performs RL on a drifting backbone by drifting fresh policy samples toward critic-ranked top- K candidates. Porygon differs in three ways: we update a residual on a frozen base rather than the full policy, we transport with an explicit pointwise trust region and a restoring anchor field (the stability contract our toy shows is load-bearing), and our reward field is a dial — including the analytic $\nabla_a Q$ source that dominates when the critic generalizes — rather than a fixed ranking surrogate.

Residual RL and staying on-manifold. Learning a residual on a frozen base is a long-standing recipe for sample-efficient robot RL; DAWN (Anonymous, 2026a) identifies the value cold-start and scale pathologies of residual critics that we also observe. The off-manifold drift our toy exhibits is the mechanism behind over-optimization and mode collapse in RL-finetuned generators (Black et al., 2024; Fan et al., 2023; Nakamoto et al., 2023), and forgetting in RL-finetuned language models is likewise governed by divergence from the base (Anonymous, 2025b) — which cannot even be measured without likelihoods, reinforcing update rules whose step size is bounded in action space by construction.

In sum, prior work either trusts the critic’s gradient and pays for it off-manifold, or discards it and pays on hard tasks; Porygon is, to our knowledge, the first fine-tuner for one-step generative policies whose update form is safe under an arbitrary critic while still admitting the full first-order signal when it is available.

3 Problem Statement

3.1 Setup

We consider an episodic decision process with states s , action chunks $a \in \mathbb{R}^{H \times d}$ (H control steps of a d -dimensional action, treated as a single vector of dimension $S = Hd$), and sparse task-success reward. A one-step generative policy maps a state and a noise sample to an action in a single network evaluation,

$$a = \pi_0(s, z), \quad z \sim \mathcal{N}(0, I), \quad (1)$$

and is pretrained on demonstrations with a drifting-model objective (Deng et al., 2026): at each step, a population of policy samples is transported by a kernel velocity field toward the data distribution, and the network is regressed onto the transported targets. We write $\mathcal{V}(x \mid p^+, p^-)$ for this transport field acting on query particles x , attracted to particles of p^+ and repelled from particles of p^- (Section 4.2 recalls its exact form).

Following residual fine-tuning practice (Anonymous, 2026a;c), we freeze the base and learn a residual head on top of it,

$$a = \pi_0(s, z) + r_\theta(s, z), \quad (2)$$

together with an ensemble critic $Q_\phi(s, z, a)$ trained by temporal difference with a conservative (CQL-style) penalty (Kumar et al., 2020). The critic training is standard and identical for every method compared in this paper; the object of study is solely the actor update — how the policy consumes the critic.

3.2 Two structural failure modes

No likelihoods. Policy-gradient fine-tuners (PPO/GRPO-style, e.g. SimpleVLA-RL (Anonymous, 2025d)) require $\log \pi(a | s)$. A one-step generator defines its action distribution implicitly through the noise map Equation (2); its likelihood is intractable, and unlike multi-step diffusion policies there is no chain-MDP reformulation to fall back on (Ren et al., 2025; Park et al., 2025; Anonymous, 2025a). The ratio-based toolbox is structurally unavailable.

Critic exploitation through backpropagation. The remaining standard update — backpropagate $-Q$ through the actor, as in DDPG lineage methods and in DICE-RL’s residual actor (Anonymous, 2026c) — couples the size of the policy update to the magnitude of the critic’s action-gradient: $\nabla_\theta \mathbb{E}[-Q] = -\nabla_a Q \cdot \partial a / \partial \theta$. A learned critic is only trained to rank actions near the data manifold; nothing constrains its gradient field off-manifold, where it can be both steep and wrong (Fujimoto et al., 2018; 2019; Anonymous, 2025c). A steep-but-wrong gradient therefore marches the policy off the data manifold at maximum speed — the better the critic looks to the optimizer, the worse the true policy becomes.

3.3 A two-regime toy experiment

We isolate the second failure mode in a bandit with a bimodal expert distribution, a one-step base pretrained with the drift operator, and a critic fit only on on-manifold data. We compare the standard backpropagation update without and with a BC anchor ($-Q$ and $-Q + \lambda \|r\|^2$, the DICE-RL residual actor) against the field-target update of Section 4, in two regimes (Figure 1; 3 seeds).

Easy regime (2-D actions, dense critic data): the critic generalizes, its gradient field is trustworthy, and backpropagation wins — true reward 0.793 ($-Q$) and 0.770 ($-Q + \text{anchor}$) versus 0.530 for a value-only field update and 0.496 for the base. When the critic is a good compass, following its gradient is the right thing to do.

Hard regime (16-D actions, 60 critic training points — mimicking high-dimensional chunked control with scarce coverage): the same backpropagation update collapses to 0.000 true reward while its internal estimate soars (Q rises by 2.9×10^4 ; 100% of actions off-manifold). Critically, the BC anchor does not save it: $-Q + \lambda \|r\|^2$ also reaches 0.000 (99% off-manifold). Penalty terms compete with an unbounded gradient and lose. The field-target update, whose displacement is clipped in action space and anchored by a restoring field, stays entirely on-manifold and holds the base’s performance (0.614 vs. base 0.647) while preserving action dispersion (best-of-16 sampling reaches 0.767).

Two conclusions set up the method. First, the failure is regime-dependent: gradient-following is optimal exactly when the critic’s gradients generalize, and catastrophic when the critic can only rank. Any fix must retain the first-order signal where it is good. Second, what protects the field update in the hard regime is not the choice of reward signal but the form of the update: displacement bounded pointwise in action space, plus a restoring anchor field — properties a backpropagated loss cannot express, since gradient clipping in parameter space bounds neither. This motivates a method that keeps the safe update form and plugs in the strongest reward field the critic supports — in slogan form: keep the compass, but bolt it to a governor.

4 Method

Porygon fine-tunes the residual head of Equation (2) with the same primitive that pretrained the base: move samples along a velocity field, then regress the network onto the moved

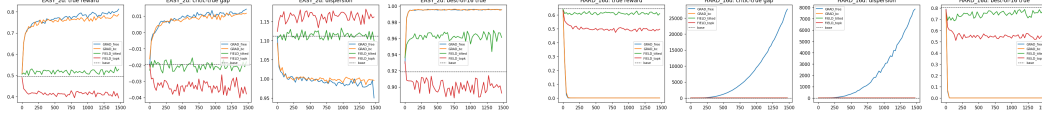


Figure 1: The two-regime toy (3 seeds). Left, easy regime (2-D actions, dense critic data): the critic’s gradient generalizes and backpropagation ($-Q$, with or without a BC anchor) is the best update. Right, hard regime (16-D actions, 60 critic points): the same updates collapse to zero true reward while their internal Q -estimate explodes — the BC anchor does not help — whereas the bounded field-target update stays on-manifold and holds the base policy. A useful fine-tuner must keep the gradient where it is trustworthy and survive where it is not.

targets. Reward shaping, trust region, and anchoring are all expressed as fields in action space; no likelihood is ever required, and no critic gradient is ever backpropagated into the actor.

4.1 The field-target update

At each update, for a state batch we draw K noise particles per state and form the current actions $a_i = \pi_0(s, z_i) + r_\theta(s, z_i)$ (detached). Each particle is displaced by a combined velocity,

$$\tilde{r}_i = \text{sg} \left[r_i + \text{clip}_{\delta_Q} (V_Q(s, z_i, a_i)) + V_{\text{anc}}(r_i) \right], \quad \mathcal{L}_{\text{actor}} = \frac{1}{K} \sum_i \|r_\theta(s, z_i) - \tilde{r}_i\|^2, \quad (3)$$

where $\text{sg}[\cdot]$ is stop-gradient and $\text{clip}_\delta(v) = v \cdot \min(1, \delta/\|v\|)$ rescales a field vector to norm at most δ . The total displacement applied to any particle is additionally clipped to δ_{tot} . Because the target is built from detached quantities, Equation (3) trains only the residual head; the critic (Section 4.4) is trained independently.

Why this primitive, for this policy class. Field-target regression is not merely a safe update — it is the operation the network was shaped by. A step-distilled generator fine-tuned under a mismatched objective drifts from the distribution its one-step map encodes (Anonymous, 2026d; Yang et al., 2024); by keeping fine-tuning inside the pretraining operator family, Porygon changes only where the targets sit, not what kind of function the network is asked to represent. We refer to this as operator-consistent fine-tuning.

A pointwise trust region in action space. The clip in Equation (3) is the load-bearing difference from backpropagation. Under $\mathcal{L} = \mathbb{E}[-Q]$, the actor’s step scales with $\|\nabla_a Q\|$ — the critic-exploitation channel of Section 3.2. Under Equation (3), the policy (not the parameters) provably moves at most δ_{tot} per update at every particle, regardless of how steep or pathological the critic is. Gradient clipping in parameter space provides no such guarantee. This is what allows Porygon to run aggressive reward steps ($\eta_Q = 0.5$) that destabilize the backpropagated control (Section 5).

4.2 The reward field V_Q

The default (and headline) reward field is the normalized critic gradient evaluated at each particle,

$$V_Q^{\text{grad}}(s, z, a) = \eta_Q \frac{\nabla_a \bar{Q}_\phi(s, z, a)}{q_{\text{scale}}}, \quad q_{\text{scale}} = \mathbb{E}[|\bar{Q}_\phi|], \quad (4)$$

where $\bar{Q}_\phi$ is the conservative ensemble aggregate. This is deliberately the same scalar the DICE-RL actor optimizes: in the small-step, unclipped limit, minimizing Equation (3) recovers the backpropagated update direction exactly (Proposition 1), so any performance difference is attributable to the update form — the action-space trust region and target regression — not to a different reward signal.

The kernel transport operator. The lower-resolution reward fields below, the pretraining objective, and the distributional anchor variant all use the same drifting-model transport operator (Deng et al., 2026). Given query particles x , attractor particles p^+ (optionally weighted), and repulsor particles p^- , distances are normalized by a global scale and, for each kernel radius R in a small set, a soft mutual-matching affinity $A = \sqrt{\text{softmax}_{\text{row}}(-d/R) \odot \text{softmax}_{\text{col}}(-d/R)}$ converts matched attractors into pulls and matched repulsors into pushes; forces are summed over radii to give $\mathcal{V}(x | p^+, p^-)$ (full form in Appendix C). Intuitively, $\mathcal{V}$ moves a particle cloud so its density flows toward p^+ and away from p^- — it needs only samples of the target, never a density or a gradient.

Because V_Q enters Equation (3) only as a velocity, the critic can be consumed at lower resolution when its gradients are not trustworthy, using only Q -values:

- zeroth-order (top- k): rank the K particles by Q and set $V_Q = \mathcal{V}(a | \text{top-}k \text{ particles, all particles})$ — the pretraining kernel transport toward the critic’s preferred candidates;
- tilted: let every particle attract with weight $K \cdot \text{softmax}_i(\text{adv}_i/\tau)$, transporting the cloud toward its own $\exp(Q/\tau)$ -tilted density, the canonical improvement target of Peng et al. (2019); Abdolmaleki et al. (2018).

These are not competing methods but settings of a dial — how much of the critic’s local structure the update trusts: full first-order information, ranking only, or soft value weights. Section 5 maps the regimes: the gradient source dominates wherever the critic generalizes, while the value-only sources close the gap precisely in the weak-critic regime predicted by Section 3.3.

4.3 The anchor field

The kernel reward fields vanish beyond their largest radius and the gradient field carries no notion of the data manifold, so Equation (3) alone would let the residual random-walk. The anchor is a restoring field with a dead zone,

$$V_{\text{anc}}(r) = -\eta_{\text{anc}} \left(1 - \frac{\rho}{\|r\|} \right)_+ r, \quad (5)$$

which is inert inside the trust radius ρ and pulls back only the excess residual norm beyond it. Unlike a $\lambda\|r\|^2$ penalty — which competes with an unbounded gradient and loses (Section 3.3) — Equation (5) composes with the clipped reward field as a velocity, so the equilibrium residual scale is set geometrically ($\|r\| \approx \rho + \delta_Q/\eta_{\text{anc}}$) rather than by a loss-weight arms race. Setting $\rho = 0$ recovers the pointwise pull $-\eta_{\text{anc}}r$, whose fixed point reproduces the DICE-RL anchor; the dead zone matters exactly when the reward field is weak (Section 5, LIBERO).

4.4 Relation to the backpropagated actor; critic

Proposition 1 (Small-step equivalence). At the current parameters, the gradient of Equation (3) is $\nabla_{\theta} \mathcal{L}_{\text{actor}} \propto -(\text{clip}_{\delta_Q}(V_Q) + V_{\text{anc}})^{\top} \partial r_{\theta} / \partial \theta$. With $V_Q = V_Q^{\text{grad}}$, no clipping ($\delta_Q, \delta_{\text{tot}} \rightarrow \infty$) and $\rho = 0$, this is the DICE-RL residual actor gradient with anchor weight η_{anc} .

Porygon is therefore a strict generalization of its control: the new degrees of freedom are the pointwise action-space clip, the dead-zone anchor, and the resolution dial on V_Q — each of which we ablate.

Remark (not just clipped gradient ascent). One might object that Equation (3) with V_Q^{grad} is “DDPG with clipping.” The distinction is where the bound lives. Clipping ∇_{θ} bounds a parameter step, whose effect on the policy output is unknown and state-dependent; Equation (3) bounds the policy’s motion at every particle, which is the quantity that off-manifold failure depends on (Section 3.3). Moreover, only the regression form makes the two other

Algorithm 1 Porygon: one update (V_Q^{grad} variant)

Require: frozen base π_0 ; residual r_θ ; critic Q_ϕ ; steps $\eta_Q, \eta_{\text{anc}}$; clips $\delta_Q, \delta_{\text{tot}}$; dead zone ρ ; particles K

- 1: sample batch of states s ; draw $z_i \sim \mathcal{N}(0, I)$, $a_i \leftarrow \pi_0(s, z_i) + r_\theta(s, z_i)$ $\triangleright$ no grad
- 2: $V_i \leftarrow \text{clip}_{\delta_Q}(\eta_Q \nabla_a \bar{Q}_\phi(s, z_i, a_i) / q_{\text{scale}})$ $\triangleright$ or top- k / tilted kernel field
- 3: $V_i \leftarrow V_i - \eta_{\text{anc}}(1 - \rho / \|r_i\|)_+ r_i$ $\triangleright$ dead-zone anchor, Equation (5)
- 4: $\tilde{r}_i \leftarrow \text{sg}[r_i + \text{clip}_{\delta_{\text{tot}}}(V_i)]$ $\triangleright$ policy moves $\leq \delta_{\text{tot}}$, always
- 5: $\theta \leftarrow \theta - \alpha \nabla_\theta \frac{1}{K} \sum_i \|r_\theta(s, z_i) - \tilde{r}_i\|^2$
- 6: update ϕ : ensemble TD + CQL penalty $\triangleright$ unchanged from Anonymous (2026c)

components expressible at all: a dead-zone anchor and a value-only V_Q have no loss-gradient counterpart. The ablations in Section 5.4 test each distinction in isolation. The critic is unchanged from DICE-RL: ensemble TD with a CQL-style penalty pushing Q down at policy actions and up at data actions; field construction never leaks gradients into the critic (gradients w.r.t. actions only). Algorithm 1 summarizes one joint update.

5 Experiments

Our experiments answer three questions. Q1: Does the field-target form beat backpropagation given the same reward signal and budget? Q2: Can a one-step drifting base fine-tuned with Porygon overtake the stronger flow-matching pipeline? Q3: When should the reward field use the critic’s gradient versus only its values — does the regime picture of Section 3.3 hold on real benchmarks?

5.1 Setup

Benchmarks. (i) robomimic can and square (state-based; 300-episode evaluations, average of the last three checkpoints; 3 seeds), run inside the official DICE-RL harness with only the actor update replaced — critic, replay, demonstrations, and evaluation protocol are byte-identical to the baseline. (ii) The eight hardest tasks of LIBERO-90 (multitask drifting base; 100 RL iterations; final numbers use powered evaluation: 100 rollouts $\times$ 3 evaluation seeds). **[TODO: appendix table: env details, horizons, demo counts, budgets.]**

Compared methods. All arms share the drifting base, residual parameterization, critic, and data; only the actor update differs:

- A (control): the DICE-RL residual actor — backpropagated $-Q$ with a filtered BC penalty (Anonymous, 2026c) — on our drifting base;
- B (Porygon): Equation (3) with the gradient reward field V_Q^{grad} ;
- C / T: the same update with the zeroth-order top- k / tilted reward fields (the resolution dial);
- FM+DICE: the unmodified state-of-the-art pipeline on the flow-matching base (Anonymous, 2026c).

Hyperparameters are family-shared across B/C/T ($\eta_Q=0.5$, $\delta_{\text{tot}}=0.15$, $\eta_{\text{anc}}=1.0$, $\tau=1$, $k=4$; dead zone $\rho=0.05$ on LIBERO) so that comparisons isolate the update rule rather than tuning effort.

5.2 Robomimic: the gradient regime

Table 1 gives the per-seed results. On can, Porygon reaches 0.97–0.99 from a base at 0.877, above both the backpropagated control on the same base (0.950) and the flow-matching pipeline (0.957) — a weaker base plus a safer update overtakes a stronger base plus the standard update (Q2). On square, the harder task, Porygon attains 0.92–0.94 on every

Table 1: robomimic success rate (300-episode evaluations, last-3-checkpoint average, per seed). “—” = matched-budget run still in progress at submission-draft time; no provisional numbers are shown.

arm	can			square		
	s42	s43	s44	s42	s43	s44
drift base (no RL)		0.877			—	
A: DICE-RL actor on drift		0.950		—	—	—
FM + DICE-RL		0.957		—	—	—
C: top- k field	0.920	0.905	0.929	0.591	0.570	0.598
T: tilted field	0.961	0.952	0.945	0.596	0.593	0.568
B: Porygon	0.988	0.993	0.966	0.923	0.938	0.929

Table 2: LIBERO hard-8 mean success (powered evaluation: 100 rollouts $\times$ 3 seeds). Runs in progress; table reserved.

	BC base	A (control)	FM + DICE	Porygon / C / T
hard-8 mean	—	—	—	—

seed. [TODO: complete Q1/Q2 square comparison sentences when the matched-budget A and FM+DICE square runs finish.]

Why does the form win? The trust region converts critic error from a stability threat into bounded noise: with policy motion capped at δ_{tot} per update, Porygon tolerates a reward step ($\eta_Q=0.5$) far above what the backpropagated control can survive, and simply takes more, safer steps in the directions where the critic is consistent. The control must instead keep its effective step small enough for its worst critic errors, which is exactly the plateau behavior we observe [TODO: quantify with the stability-window ablation (Section 5.4, iii) and the matched-budget A square endpoint].

The dial answers Q3 in its first regime: with a critic that generalizes (robomimic critics see dense, single-task data), the value-only reward fields preserve can (0.90–0.96) but collapse on square (~ 0.6 versus Porygon’s ~ 0.93), despite verified-healthy residuals and longer training. Ranking resolution throws away exactly the signal that solves the harder task — the real-benchmark mirror of the toy’s easy regime, where the compass wins.

5.3 LIBERO hard-8: the weak-critic regime

The multitask LIBERO setting flips the critic’s situation: eight hard tasks share one critic conditioned on scarcer per-task data, and our diagnostics show its action-gradient carries far less signal ([TODO: cite diagnostic numbers: $\text{SNR}_{\nabla Q}$ from the field viability probe]). The toy predicts the dial should compress: value-only transport should approach — not trail — the gradient field.

[TODO: Fill Table 2 and write the compression analysis when the chains reach iteration 100 and powered evals complete. Include the guarded BC-filter interaction.]

5.4 Ablations: which component pays where

[TODO: Four ablations, one table each once runs complete: (i) trust-region locus — control A + parameter-space gradient clipping vs. B’s action-space clip at equal effective step (isolates the central claim); (ii) anchor geometry — $\rho=0$ silently erases the residual under weak reward fields (measured residual RMS collapses $\sim 5\times$ and performance reverts to base-plus-selection), the quantitative version of the toy’s “penalties lose to fields” lesson; (iii) step/clip sweep — the 21-arm square study behind the family recipe: stability window of η_Q under Equation (3) vs. backpropagation; (iv) dispersion retention — action dispersion and best-of- N gains after fine-tuning, vs. control (does the field update preserve the multimodality the base was trained for?).]

6 Conclusion

We showed that the natural update primitives for RL fine-tuning of one-step generative policies — likelihood ratios and backpropagated critics — are respectively unavailable and unsafe, and that the policy class’s own pretraining primitive, field-target regression, provides the missing update rule. Porygon expresses reward, trust region, and anchoring as composable velocity fields over action particles, provably bounds policy movement pointwise in action space, and recovers the backpropagated direction in the small-step limit. On robomimic it lifts a drifting base past the stronger flow-matching + DICE-RL pipeline; a controlled toy and a weak-critic benchmark map when the critic’s gradient, ranking, or soft values should drive the field.

Limitations. On the weak-critic benchmark Porygon is competitive with, but not yet ahead of, the flow-matching pipeline [TODO: update when final]; our critic-trust diagnosis is post hoc — choosing the reward-field resolution online from critic trust signals is the natural next step; and all results are state-based simulation benchmarks on a single embodiment. The broader principle — fine-tune a policy with the operator that pretrained it — applies to any generative policy class trained by target regression, and testing it beyond drifting models is the direction we find most promising.

Reproducibility statement

[TODO: Code release (training framework + official-harness implant), exact configs (family-shared recipe), seeds, eval protocols (300-ep robomimic; 100×3 powered LIBERO), compute (single L40S/A100 per run, 8–20h), toy experiment script. All numbers in Table 1 are averages, never best-of.]

References

- Abbas Abdolmaleki, Jost Tobias Springenberg, Yuval Tassa, Remi Munos, Nicolas Heess, and Martin Riedmiller. Maximum a posteriori policy optimisation. In International Conference on Learning Representations (ICLR), 2018.
- Anonymous. Reinflow: Fine-tuning flow matching policy with online reinforcement learning. arXiv preprint arXiv:2505.22094, 2025a.
- Anonymous. RL’s razor: Why online reinforcement learning forgets less. arXiv preprint arXiv:2509.04259, 2025b.
- Anonymous. Successive action and value optimization. Reinforcement Learning Conference (RLC), 2025c.
- Anonymous. Simplevla-rl: Scaling vla training via reinforcement learning. arXiv preprint, 2025d.
- Anonymous. Dawn: Warm-starting residual reinforcement learning. arXiv preprint arXiv:2602.10539, 2026a.
- Anonymous. Drifting field policy. arXiv preprint arXiv:2605.07727, 2026b.
- Anonymous. Dice-rl: Residual reinforcement fine-tuning of one-step distilled policies. In International Conference on Machine Learning (ICML), 2026c. arXiv:2603.10263.
- Anonymous. On the objective mismatch of fine-tuning step-distilled generators. arXiv preprint arXiv:2605.05204, 2026d.
- Kevin Black, Michael Janner, Yilun Du, Ilya Kostrikov, and Sergey Levine. Training diffusion models with reinforcement learning. In International Conference on Learning Representations (ICLR), 2024. arXiv:2305.13301.
- Deng, He, et al. Drifting models: One-step generative modeling via sample transport. arXiv preprint arXiv:2602.04770, 2026.

- Ying Fan, Olivia Watkins, Yuqing Du, Hao Liu, Moonkyung Ryu, Craig Boutilier, Pieter Abbeel, Mohammad Ghavamzadeh, Kangwook Lee, and Kimin Lee. DPOK: Reinforcement learning for fine-tuning text-to-image diffusion models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Scott Fujimoto, Herke van Hoof, and David Meger. Addressing function approximation error in actor-critic methods. In *International Conference on Machine Learning (ICML)*, 2018.
- Scott Fujimoto, David Meger, and Doina Precup. Off-policy deep reinforcement learning without exploration. In *International Conference on Machine Learning (ICML)*, 2019.
- Tuomas Haarnoja, Haoran Tang, Pieter Abbeel, and Sergey Levine. Reinforcement learning with deep energy-based policies. In *International Conference on Machine Learning (ICML)*, 2017.
- Philippe Hansen-Estruch, Ilya Kostrikov, Michael Janner, Jakub Grudzien Kuba, and Sergey Levine. IDQL: Implicit Q-learning as an actor-critic method with diffusion policies. *arXiv preprint arXiv:2304.10573*, 2023.
- Dmitry Kalashnikov, Alex Irpan, Peter Pastor, Julian Ibarz, Alexander Herzog, Eric Jang, Deirdre Quillen, Ethan Holly, Mrinal Kalakrishnan, Vincent Vanhoucke, and Sergey Levine. QT-Opt: Scalable deep reinforcement learning for vision-based robotic manipulation. In *Conference on Robot Learning (CoRL)*, 2018.
- Aviral Kumar, Aurick Zhou, George Tucker, and Sergey Levine. Conservative Q-learning for offline reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- Yang Liu, Prajit Ramachandran, Qiang Liu, and Jian Peng. Stein variational policy gradient. *arXiv preprint arXiv:1704.02399*, 2017.
- Max Sobol Mark et al. Policy-agnostic RL: Offline RL and online RL fine-tuning of any class and backbone. *arXiv preprint arXiv:2412.06685*, 2024.
- Mitsuhiko Nakamoto, Yuexiang Zhai, Anikait Singh, Max Sobol Mark, Yi Ma, Chelsea Finn, Aviral Kumar, and Sergey Levine. Cal-QL: Calibrated offline RL pre-training for efficient online fine-tuning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Seohong Park, Qiyang Li, and Sergey Levine. Flow Q-learning. *arXiv preprint arXiv:2502.02538*, 2025.
- Xue Bin Peng, Aviral Kumar, Grace Zhang, and Sergey Levine. Advantage-weighted regression: Simple and scalable off-policy reinforcement learning. *arXiv preprint arXiv:1910.00177*, 2019.
- Allen Z. Ren, Justin Lidard, Lars L. Ankile, Anthony Simeonov, Pulkit Agrawal, Anirudha Majumdar, Benjamin Burchfiel, Hongkai Dai, and Max Simchowitz. Diffusion policy optimization. In *International Conference on Learning Representations (ICLR)*, 2025. *arXiv:2409.00588*.
- Ziyu Wang, Alexander Novikov, Konrad Zolna, Josh Merel, Jost Tobias Springenberg, Scott Reed, Bobak Shahriari, Noah Siegel, Caglar Gulcehre, Nicolas Heess, and Nando de Freitas. Critic regularized regression. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- Zhaorui Yang, Tianyu Pang, Haozhe Feng, Han Wang, Wei Chen, Minfeng Zhu, and Qian Liu. Self-distillation bridges distribution gap in language model fine-tuning. *arXiv preprint arXiv:2402.13669*, 2024.

A Additional results

[TODO: Full seed tables, training curves, hard-8 per-task breakdown.]

B Hyperparameters

[TODO: Family-shared recipe table; official DICE-RL hypers; sweep grids.]

C The kernel transport operator

[TODO: Full form of $\mathcal{V}$: global distance scale, per-radius softmax-softmax affinity, contested-attractor weighting, self-exclusion, per-radius normalization on/off (restoring vs. constant-pull), radii $\{0.02, 0.05, 0.2\}$. Port from drift_field.py docstrings.]

D Toy two-regime construction

[TODO: Details of the hard-critic/easy-critic toy; failure of compass updates.]